

The Abstraction Gap Between Mathematics and Physics: The Hidden Compromise in Dimensional Analysis Resolved

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Abstract

In this dataset we bridge the abstraction gap between mathematics and physics by adopting a geometric perspective of trigonometry. We formalize this by addressing the arc length of a circular sector and defining the circular functions axiomatically as direct functional mappings from the physical parameters of a circular sector – namely the radius and the arc length – to its orthogonal projections. Finally, we demonstrate the utility of this framework by modeling simple harmonic motion and uniform circular motion through these geometric functions.

0.1 Circular Sector

0.1.1 Arc Length

$$\begin{aligned}
 a_0 &= x, -r \leq x \leq r \\
 a_n &= \frac{r\sqrt{r^2 - a_{n-1}^2}}{\sqrt{r^2 + r^2 - 2ra_{n-1}}}, n \geq 1 \\
 a_n &= \frac{r\sqrt{r^2 - a_{n-1}^2}}{\sqrt{2r^2 - 2ra_{n-1}}}, n \geq 1 \\
 a_n &= r \frac{\sqrt{r^2 - a_{n-1}^2}}{\sqrt{2r^2 - 2ra_{n-1}}}, n \geq 1 \\
 a_n &= r \sqrt{\frac{r^2 - a_{n-1}^2}{2r^2 - 2ra_{n-1}}}, n \geq 1 \\
 a_n &= r \sqrt{\frac{(r + a_{n-1})(r - a_{n-1})}{2r^2 - 2ra_{n-1}}}, n \geq 1 \\
 a_n &= r \sqrt{\frac{(r + a_{n-1})(r - a_{n-1})}{2r(r - a_{n-1})}}, n \geq 1 \\
 a_n &= r \sqrt{\frac{r + a_{n-1}}{2r}}, n \geq 1 \\
 a_n &= r \sqrt{\frac{2r(r + a_{n-1})}{(2r)^2}}, n \geq 1 \\
 a_n &= r \sqrt{\frac{2r^2 + 2ra_{n-1}}{(2r)^2}}, n \geq 1 \\
 a_n &= r \sqrt{\frac{\left(\sqrt{2r^2 + 2ra_{n-1}}\right)^2}{(2r)^2}}, n \geq 1 \\
 a_n &= r \sqrt{\left(\frac{\sqrt{2r^2 + 2ra_{n-1}}}{2r}\right)^2}, n \geq 1 \\
 a_n &= r \frac{\sqrt{2r^2 + 2ra_{n-1}}}{2r}, n \geq 1 \\
 a_n &= \frac{r\sqrt{2r^2 + 2ra_{n-1}}}{2r}, n \geq 1 \\
 a_n &= \frac{\sqrt{2r^2 + 2ra_{n-1}}}{2}, n \geq 1 \\
 a_n &= \frac{\sqrt{2r^2 + 2ra_{n-1}}}{2}, n \geq 1 \\
 a_0 &= x, -r \leq x \leq r \\
 s &= \lim_{n \rightarrow \infty} 2^n \sqrt{r^2 + r^2 - 2ra_n} \\
 a_n &= \frac{\sqrt{2r^2 + 2ra_{n-1}}}{2}, n \geq 1 \\
 a_0 &= x, -r \leq x \leq r \\
 s &= \lim_{n \rightarrow \infty} 2^n \sqrt{2r^2 - 2ra_n} \\
 a_n &= \frac{\sqrt{2r^2 + 2ra_{n-1}}}{2}, n \geq 1 \\
 a_0 &= x, -r \leq x \leq r
 \end{aligned}$$

0.1.2 Area

$$A = \frac{rs}{2}$$

0.1.3 Circular Functions

0.1.3.1 Arccosine

$$\begin{aligned} \arccos: \{(r, x) \in \mathbb{R}^2 \mid r > 0, -r \leq x \leq r\} &\rightarrow \mathbb{R} \\ (r, x) &\mapsto s \end{aligned}$$

$$\begin{aligned} \arccos(r, x) &= \lim_{n \rightarrow \infty} 2^n \sqrt{2r^2 - 2ra_n} \\ a_n &= \frac{\sqrt{2r^2 + 2ra_{n-1}}}{2}, n \geq 1 \\ a_0 &= x \end{aligned}$$

0.1.3.2 Cosine

$$\begin{aligned} \cos: \{(r, s) \in \mathbb{R}^2 \mid r > 0\} &\rightarrow \mathbb{R} \\ (r, s) &\mapsto x \end{aligned}$$

$$s = \lim_{n \rightarrow \infty} 2^n \sqrt{2r^2 - 2ra_n}$$

$$a_n = \frac{\sqrt{2r^2 + 2ra_{n-1}}}{2}, n \geq 1$$

$$a_0 = \cos(r, s)$$

$$s = \lim_{m \rightarrow \infty} 2^m \sqrt{2r^2 - 2ra_{m,m}}$$

$$a_{m,n} = \frac{\sqrt{2r^2 + 2ra_{m,n-1}}}{2}, n \geq 1, n \leq m$$

$$a_{m,0} = \cos(r, s), m \geq 0$$

$$\cos(r, s) = \lim_{m \rightarrow \infty} a_{m,m}$$

$$a_{m,n-1} = \frac{\sqrt{2r^2 + 2ra_{m,n}}}{2}, n \geq 1, n \leq m$$

$$2^m \sqrt{2r^2 - 2ra_{m,0}} = s, m \geq 0$$

$$a_{m,n-1} = \frac{\sqrt{2r^2 + 2ra_{m,n}}}{2}, n \geq 1, n \leq m$$

$$2^m \sqrt{2r^2 - 2ra_{m,0}} = s, m \geq 0$$

$$a_{m,n-1} = \frac{\sqrt{2r^2 + 2ra_{m,n}}}{2}, n \geq 1, n \leq m$$

$$2a_{m,n-1} = 2 \frac{\sqrt{2r^2 + 2ra_{m,n}}}{2}, n \geq 1, n \leq m$$

$$2a_{m,n-1} = \frac{2\sqrt{2r^2 + 2ra_{m,n}}}{2}, n \geq 1, n \leq m$$

$$2a_{m,n-1} = \sqrt{2r^2 + 2ra_{m,n}}, n \geq 1, n \leq m$$

$$(2a_{m,n-1})^2 = \left(\sqrt{2r^2 + 2ra_{m,n}} \right)^2, n \geq 1, n \leq m$$

$$2^2 a_{m,n-1}^2 = \left(\sqrt{2r^2 + 2ra_{m,n}} \right)^2, n \geq 1, n \leq m$$

$$4a_{m,n-1}^2 = \left(\sqrt{2r^2 + 2ra_{m,n}} \right)^2, n \geq 1, n \leq m$$

$$4a_{m,n-1}^2 = 2r^2 + 2ra_{m,n}, n \geq 1, n \leq m$$

$$4a_{m,n-1}^2 - 2r^2 = 2r^2 + 2ra_{m,n} - 2r^2, n \geq 1, n \leq m$$

$$4a_{m,n-1}^2 - 2r^2 = 2ra_{m,n}, n \geq 1, n \leq m$$

$$\frac{4a_{m,n-1}^2 - 2r^2}{2r} = \frac{2ra_{m,n}}{2r}, n \geq 1, n \leq m$$

$$\frac{4a_{m,n-1}^2}{2r} - \frac{2r^2}{2r} = \frac{2ra_{m,n}}{2r}, n \geq 1, n \leq m$$

$$\frac{2a_{m,n-1}^2}{r} - \frac{2r^2}{2r} = \frac{2ra_{m,n}}{2r}, n \geq 1, n \leq m$$

$$\frac{2a_{m,n-1}^2}{r} - r = \frac{2ra_{m,n}}{2r}, n \geq 1, n \leq m$$

$$\frac{2a_{m,n-1}^2}{r} - r = a_{m,n}, n \geq 1, n \leq m$$

$$a_{m,n} = \frac{2a_{m,n-1}^2}{r} - r, n \geq 1, n \leq m$$

$$\frac{2^m \sqrt{2r^2 - 2ra_{m,0}}}{2^m} = \frac{s}{2^m}, m \geq 0$$

$$\sqrt{2r^2 - 2ra_{m,0}} = \frac{s}{2^m}, m \geq 0$$

$$\left(\sqrt{2r^2 - 2ra_{m,0}} \right)^2 = \left(\frac{s}{2^m} \right)^2, m \geq 0$$

$$2r^2 - 2ra_{m,0} = \left(\frac{s}{2^m} \right)^2, m \geq 0$$

$$2r^2 - 2ra_{m,0} = \frac{s^2}{(2^m)^2}, m \geq 0$$

$$2r^2 - 2ra_{m,0} = \frac{s^2}{2^{2m}}, m \geq 0$$

$$2r^2 - 2ra_{m,0} - 2r^2 = \frac{s^2}{2^{2m}} - 2r^2, m \geq 0$$

$$-2ra_{m,0} = \frac{s^2}{2^{2m}} - 2r^2, m \geq 0$$

$$\frac{-2ra_{m,0}}{-2r} = \frac{\left(\frac{s^2}{2^{2m}} \right) - 2r^2}{-2r}, m \geq 0$$

$$a_{m,0} = \frac{\left(\frac{s^2}{2^{2m}} \right) - 2r^2}{-2r}, m \geq 0$$

$$a_{m,0} = \frac{\left(\frac{s^2}{2^{2m}} \right)}{-2r} - \frac{2r^2}{-2r}, m \geq 0$$

$$a_{m,0} = \frac{s^2}{2^{2m}(-2r)} - \frac{2r^2}{-2r}, m \geq 0$$

$$a_{m,0} = \frac{s^2}{-2^{2m+1}r} - \frac{2r^2}{-2r}, m \geq 0$$

$$a_{m,0} = \frac{s^2}{-2^{2m+1}r} + \frac{-2r^2}{-2r}, m \geq 0$$

$$a_{m,0} = \frac{s^2}{-2^{2m+1}r} + r, m \geq 0$$

$$a_{m,0} = r + \frac{s^2}{-2^{2m+1}r}, m \geq 0$$

$$a_{m,0} = r - \left(-\frac{s^2}{-2^{2m+1}r} \right), m \geq 0$$

$$a_{m,0} = r - \left(\frac{-s^2}{-2^{2m+1}r} \right), m \geq 0$$

$$a_{m,n} = \frac{2a_{m,n-1}^2}{r} - r, n \geq 1, n \leq m$$

$$a_{m,0} = r - \frac{s^2}{2^{2m+1}r}, m \geq 0$$

$$r\hat{a}_{m,n} = \frac{2(r\hat{a}_{m,n-1})^2}{r} - r, n \geq 1, n \leq m$$

$$r\hat{a}_{m,0} = r - \frac{s^2}{2^{2m+1}r}, m \geq 0$$

$$r\hat{a}_{m,n} = \frac{2(r\hat{a}_{m,n-1})^2}{r} - r, n \geq 1, n \leq m$$

$$r\hat{a}_{m,n} = \frac{2r^2\hat{a}_{m,n-1}^2}{r} - r, n \geq 1, n \leq m$$

$$r\hat{a}_{m,n} = 2r\hat{a}_{m,n-1}^2 - r, n \geq 1, n \leq m$$

$$\frac{r\hat{a}_{m,n}}{r} = \frac{2r\hat{a}_{m,n-1}^2 - r}{r}, n \geq 1, n \leq m$$

$$\hat{a}_{m,n} = \frac{2r\hat{a}_{m,n-1}^2 - r}{r}, n \geq 1, n \leq m$$

$$\hat{a}_{m,n} = \frac{2r\hat{a}_{m,n-1}^2}{r} - \frac{r}{r}, n \geq 1, n \leq m$$

$$\hat{a}_{m,n} = 2\hat{a}_{m,n-1}^2 - \frac{r}{r}, n \geq 1, n \leq m$$

$$\hat{a}_{m,n} = 2\hat{a}_{m,n-1}^2 - 1, n \geq 1, n \leq m$$

$$r\hat{a}_{m,0} = r - \frac{s^2}{2^{2m+1}r}, m \geq 0$$

$$\frac{r\hat{a}_{m,0}}{r} = \frac{r - \left(\frac{s^2}{2^{2m+1}r} \right)}{r}, m \geq 0$$

$$\hat{a}_{m,0} = \frac{r - \left(\frac{s^2}{2^{2m+1}r} \right)}{r}, m \geq 0$$

$$\hat{a}_{m,0} = \frac{r}{r} - \frac{\left(\frac{s^2}{2^{2m+1}r} \right)}{r}, m \geq 0$$

$$\hat{a}_{m,0} = 1 - \frac{\left(\frac{s^2}{2^{2m+1}r} \right)}{r}, m \geq 0$$

$$\hat{a}_{m,0} = 1 - \frac{s^2}{2^{2m+1}r^2}, m \geq 0$$

$$\hat{a}_{m,n} = 2\hat{a}_{m,n-1}^2 - 1, n \geq 1, n \leq m$$

$$\hat{a}_{m,0} = 1 - \frac{s^2}{2^{2m+1}r^2}, m \geq 0$$

$$\hat{a}_{m,n} = T_2^{\circ n}(\hat{a}_{m,0}), 0 \leq n \leq m$$

$$\hat{a}_{m,n} = T_2^{\circ n} \left(1 - \frac{s^2}{2^{2m+1}r^2} \right), 0 \leq n \leq m$$

$$\begin{aligned}
\hat{a}_{m,n} &= T_{2^n} \left(1 - \frac{s^2}{2^{2m+1}r^2} \right), 0 \leq n \leq m \\
\hat{a}_{m,n} &= \sum_{k=0}^{2^n} \frac{T_{2^n}^{(k)}(1)}{k!} \left(\left(1 - \frac{s^2}{2^{2m+1}r^2} \right) - 1 \right)^k, 0 \leq n \leq m \\
\hat{a}_{m,n} &= \sum_{k=0}^{2^n} \frac{T_{2^n}^{(k)}(1)}{k!} \left(1 - \frac{s^2}{2^{2m+1}r^2} - 1 \right)^k, 0 \leq n \leq m \\
\hat{a}_{m,n} &= \sum_{k=0}^{2^n} \frac{T_{2^n}^{(k)}(1)}{k!} \left(-\frac{s^2}{2^{2m+1}r^2} \right)^k, 0 \leq n \leq m \\
\hat{a}_{m,n} &= \sum_{k=0}^{2^n} \frac{2^k \prod_{j=0}^{k-1} ((2^n)^2 - j^2)}{(2k)!} \left(-\frac{s^2}{2^{2m+1}r^2} \right)^k, 0 \leq n \leq m \\
\hat{a}_{m,n} &= \sum_{k=0}^{2^n} \frac{2^k \left(-\frac{s^2}{2^{2m+1}r^2} \right)^k \prod_{j=0}^{k-1} ((2^n)^2 - j^2)}{(2k)!}, 0 \leq n \leq m \\
\hat{a}_{m,n} &= \sum_{k=0}^{2^n} \frac{\left(2 \left(-\frac{s^2}{2^{2m+1}r^2} \right) \right)^k \prod_{j=0}^{k-1} ((2^n)^2 - j^2)}{(2k)!}, 0 \leq n \leq m \\
\hat{a}_{m,n} &= \sum_{k=0}^{2^n} \frac{\left(-2 \frac{s^2}{2^{2m+1}r^2} \right)^k \prod_{j=0}^{k-1} ((2^n)^2 - j^2)}{(2k)!}, 0 \leq n \leq m \\
\hat{a}_{m,n} &= \sum_{k=0}^{2^n} \frac{\left(-\frac{2s^2}{2^{2m+1}r^2} \right)^k \prod_{j=0}^{k-1} ((2^n)^2 - j^2)}{(2k)!}, 0 \leq n \leq m \\
\hat{a}_{m,n} &= \sum_{k=0}^{2^n} \frac{\left(-\frac{s^2}{2^{2m}r^2} \right)^k \prod_{j=0}^{k-1} ((2^n)^2 - j^2)}{(2k)!}, 0 \leq n \leq m \\
\hat{a}_{m,n} &= \sum_{k=0}^{2^n} \frac{\left(-\frac{s^2}{(2^2)^m r^2} \right)^k \prod_{j=0}^{k-1} ((2^n)^2 - j^2)}{(2k)!}, 0 \leq n \leq m \\
\hat{a}_{m,n} &= \sum_{k=0}^{2^n} \frac{\left(-\frac{s^2}{4^m r^2} \right)^k \prod_{j=0}^{k-1} ((2^n)^2 - j^2)}{(2k)!}, 0 \leq n \leq m \\
\hat{a}_{m,n} &= \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{s^2}{4^m r^2} \right)^k \prod_{j=0}^{k-1} ((2^n)^2 - j^2)}{(2k)!}, 0 \leq n \leq m \\
\hat{a}_{m,n} &= \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{\left(\frac{s^2}{r^2} \right)}{4^m} \right)^k \prod_{j=0}^{k-1} ((2^n)^2 - j^2)}{(2k)!}, 0 \leq n \leq m \\
\hat{a}_{m,n} &= \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{\left(\frac{s}{r} \right)^2}{4^m} \right)^k \prod_{j=0}^{k-1} ((2^n)^2 - j^2)}{(2k)!}, 0 \leq n \leq m
\end{aligned}$$

$$\begin{aligned}
\hat{a}_{m,n} &= \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{s}{r}\right)^k \prod_{j=0}^{k-1} ((2^n)^2 - j^2)}{(2k)!}, 0 \leq n \leq m \\
\hat{a}_{m,n} &= \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k} \prod_{j=0}^{k-1} ((2^n)^2 - j^2)}{(2k)!}, 0 \leq n \leq m \\
\hat{a}_{m,n} &= \sum_{k=0}^{2^n} \frac{\left(\frac{s}{r}\right)^{2k} \prod_{j=0}^{k-1} ((2^n)^2 - j^2)}{(2k)!}, 0 \leq n \leq m \\
\hat{a}_{m,n} &= \sum_{k=0}^{2^n} \frac{\left(\frac{(-1)^k \left(\frac{s}{r}\right)^{2k} \prod_{j=0}^{k-1} ((2^n)^2 - j^2)}{(4^m)^k} \right)}{(2k)!}, 0 \leq n \leq m \\
\hat{a}_{m,n} &= \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k} \prod_{j=0}^{k-1} ((2^n)^2 - j^2)}{(4^m)^k (2k)!}, 0 \leq n \leq m \\
\hat{a}_{m,n} &= \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k} \prod_{j=0}^{k-1} ((2^n)^2 - j^2)}{(2k)! (4^m)^k}, 0 \leq n \leq m \\
\hat{a}_{m,n} &= \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k} \prod_{j=0}^{k-1} ((2^n)^2 - j^2)}{(2k)!} \frac{\prod_{j=0}^{k-1} ((2^n)^2 - j^2)}{(4^m)^k}, 0 \leq n \leq m \\
\hat{a}_{m,n} &= \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k} \prod_{j=0}^{k-1} ((2^n)^2 - j^2)}{(2k)!} \frac{\prod_{j=0}^{k-1} ((2^n)^2 - j^2)}{\prod_{j=0}^{k-1} 4^m}, 0 \leq n \leq m \\
\hat{a}_{m,n} &= \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k} \prod_{j=0}^{k-1} ((2^n)^2 - j^2)}{(2k)!} \prod_{j=0}^{k-1} \frac{(2^n)^2 - j^2}{4^m}, 0 \leq n \leq m \\
\hat{a}_{m,n} &= \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k} \prod_{j=0}^{k-1} ((2^n)^2 - j^2)}{(2k)!} \prod_{j=0}^{k-1} \frac{(2^n)^2 - j^2}{4^m}, 0 \leq n \leq m \\
\hat{a}_{m,n} &= \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k} \prod_{j=0}^{k-1} ((2^n)^2 - j^2)}{(2k)!} \prod_{j=0}^{k-1} \frac{4^n - j^2}{4^m}, 0 \leq n \leq m \\
\hat{a}_{m,n} &= \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k} \prod_{j=0}^{k-1} ((2^n)^2 - j^2)}{(2k)!} \prod_{j=0}^{k-1} \left(\frac{4^n}{4^m} - \frac{j^2}{4^m} \right), 0 \leq n \leq m \\
\hat{a}_{m,n} &= \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k} \prod_{j=0}^{k-1} ((2^n)^2 - j^2)}{(2k)!} \prod_{j=0}^{k-1} \left(4^{n-m} - \frac{j^2}{4^m} \right), 0 \leq n \leq m \\
r\hat{a}_{m,n} &= r \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k} \prod_{j=0}^{k-1} ((2^n)^2 - j^2)}{(2k)!} \prod_{j=0}^{k-1} \left(4^{n-m} - \frac{j^2}{4^m} \right), 0 \leq n \leq m \\
a_{m,n} &= r \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k} \prod_{j=0}^{k-1} ((2^n)^2 - j^2)}{(2k)!} \prod_{j=0}^{k-1} \left(4^{n-m} - \frac{j^2}{4^m} \right), 0 \leq n \leq m
\end{aligned}$$

$$\begin{aligned}
\cos(r, s) &= \lim_{m \rightarrow \infty} a_{m,m} \\
\cos(r, s) &= \lim_{m \rightarrow \infty} r \sum_{k=0}^{2^m} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!} \prod_{j=0}^{k-1} \left(4^{m-m} - \frac{j^2}{4^m}\right) \\
\cos(r, s) &= \lim_{m \rightarrow \infty} r \sum_{k=0}^{2^m} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!} \prod_{j=0}^{k-1} \left(4^0 - \frac{j^2}{4^m}\right) \\
\cos(r, s) &= \lim_{m \rightarrow \infty} r \sum_{k=0}^{2^m} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!} \prod_{j=0}^{k-1} \left(1 - \frac{j^2}{4^m}\right) \\
\cos(r, s) &= r \lim_{m \rightarrow \infty} \sum_{k=0}^{2^m} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!} \prod_{j=0}^{k-1} \left(1 - \frac{j^2}{4^m}\right) \\
\cos(r, s) &= r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!}
\end{aligned}$$

0.1.3.2.1 Partial Derivative

$$\begin{aligned}
\frac{\partial \cos(r, s)}{\partial s} &= \frac{\partial \left(r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!} \right)}{\partial s} \\
\frac{\partial \cos(r, s)}{\partial s} &= \frac{\partial \left(r \sum_{k=0}^{\infty} \frac{(-1)^k \frac{s^{2k}}{r^{2k}}}{(2k)!} \right)}{\partial s} \\
\frac{\partial \cos(r, s)}{\partial s} &= \frac{\partial \left(r \sum_{k=0}^{\infty} \frac{(-1)^k s^{2k}}{r^{2k} (2k)!} \right)}{\partial s} \\
\frac{\partial \cos(r, s)}{\partial s} &= \frac{\partial \left(\sum_{k=0}^{\infty} \frac{(-1)^k r s^{2k}}{r^{2k} (2k)!} \right)}{\partial s} \\
\frac{\partial \cos(r, s)}{\partial s} &= \frac{\partial \left(\sum_{k=0}^{\infty} \frac{(-1)^k s^{2k}}{r^{2k-1} (2k)!} \right)}{\partial s} \\
\frac{\partial \cos(r, s)}{\partial s} &= \sum_{k=0}^{\infty} \frac{(-1)^k \frac{\partial s^{2k}}{\partial s}}{r^{2k-1} (2k)!} \\
\frac{\partial \cos(r, s)}{\partial s} &= \sum_{k=1}^{\infty} \frac{(-1)^k (2k) s^{2k-1}}{r^{2k-1} (2k)!} \\
\frac{\partial \cos(r, s)}{\partial s} &= \sum_{k=1}^{\infty} \frac{(-1)^k (2k) \frac{s^{2k-1}}{r^{2k-1}}}{(2k)!} \\
\frac{\partial \cos(r, s)}{\partial s} &= \sum_{k=1}^{\infty} \frac{(-1)^k (2k) \left(\frac{s}{r}\right)^{2k-1}}{(2k)!} \\
\frac{\partial \cos(r, s)}{\partial s} &= \sum_{k=1}^{\infty} \frac{(-1)^k (2k) \left(\frac{s}{r}\right)^{2k-1}}{(2k)(2k-1)!} \\
\frac{\partial \cos(r, s)}{\partial s} &= \sum_{k=1}^{\infty} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k-1}}{(2k-1)!}
\end{aligned}$$

$$\begin{aligned}
\frac{\partial \cos(r, s)}{\partial s} &= \sum_{k=1}^{\infty} \frac{-(-1)^{k-1} \left(\frac{s}{r}\right)^{2k-1}}{(2k-1)!} \\
\frac{\partial \cos(r, s)}{\partial s} &= - \sum_{k=1}^{\infty} \frac{(-1)^{k-1} \left(\frac{s}{r}\right)^{2k-1}}{(2k-1)!} \\
\frac{\partial \cos(r, s)}{\partial s} &= - \sum_{k=1}^{\infty} \frac{(-1)^{k-1} \left(\frac{s}{r}\right)^{2(k-1)+1}}{(2k-1)!} \\
\frac{\partial \cos(r, s)}{\partial s} &= - \sum_{k=1}^{\infty} \frac{(-1)^{k-1} \left(\frac{s}{r}\right)^{2(k-1)+1}}{(2(k-1)+1)!} \\
\frac{\partial \cos(r, s)}{\partial s} &= - \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k+1}}{(2k+1)!} \\
\frac{\partial \cos(r, s)}{\partial s} &= \frac{-r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k+1}}{(2k+1)!}}{r} \\
\frac{\partial \cos(r, s)}{\partial s} &= \frac{-\sin(r, s)}{r}
\end{aligned}$$

0.1.3.3 Arcsine

$$\begin{aligned}
\arcsin: \{(r, y) \in \mathbb{R}^2 \mid r > 0, -r \leq y \leq r\} &\rightarrow \mathbb{R} \\
(r, y) &\mapsto s
\end{aligned}$$

$$\begin{aligned}
\arcsin(r, y) &= \frac{\arcsin(r, y) + \arcsin(r, y)}{2} \\
\arcsin(r, y) &= \frac{\arcsin(r, y) + \arcsin(r, y) + \arccos(r, y) - \arccos(r, y)}{2} \\
\arcsin(r, y) &= \frac{\arcsin(r, y) + \arcsin(r, -y) + \arccos(r, -y) - \arccos(r, y)}{2} \\
\arcsin(r, y) &= \frac{\arcsin(r, y) - \arcsin(r, y) + \arccos(r, -y) - \arccos(r, y)}{2} \\
\arcsin(r, y) &= \frac{\arccos(r, -y) - \arccos(r, y)}{2}
\end{aligned}$$

0.1.3.4 Sine

$$\begin{aligned}
\sin: \{(r, s) \in \mathbb{R}^2 \mid r > 0\} &\rightarrow \mathbb{R} \\
(r, s) &\mapsto y
\end{aligned}$$

$$\begin{aligned}
\sin^2(r, \alpha - \beta) + (r - \cos(r, \alpha - \beta))^2 &= (\cos(r, \alpha) - \cos(r, \beta))^2 + (\sin(r, \alpha) - \sin(r, \beta))^2 \\
\cos^2(r, \alpha - \beta) + \sin^2(r, \alpha - \beta) + (r - \cos(r, \alpha - \beta))^2 - \cos^2(r, \alpha - \beta) &= (\cos(r, \alpha) - \cos(r, \beta))^2 + (\sin(r, \alpha) - \sin(r, \beta))^2 \\
r^2 + (r - \cos(r, \alpha - \beta))^2 - \cos^2(r, \alpha - \beta) &= (\cos(r, \alpha) - \cos(r, \beta))^2 + (\sin(r, \alpha) - \sin(r, \beta))^2 \\
r^2 + r^2 - 2r \cos(r, \alpha - \beta) + \cos^2(r, \alpha - \beta) - \cos^2(r, \alpha - \beta) &= (\cos(r, \alpha) - \cos(r, \beta))^2 + (\sin(r, \alpha) - \sin(r, \beta))^2 \\
2r^2 - 2r \cos(r, \alpha - \beta) + \cos^2(r, \alpha - \beta) - \cos^2(r, \alpha - \beta) &= (\cos(r, \alpha) - \cos(r, \beta))^2 + (\sin(r, \alpha) - \sin(r, \beta))^2 \\
2r^2 - 2r \cos(r, \alpha - \beta) &= (\cos(r, \alpha) - \cos(r, \beta))^2 + (\sin(r, \alpha) - \sin(r, \beta))^2 \\
2r^2 - 2r \cos(r, \alpha - \beta) - 2r^2 &= (\cos(r, \alpha) - \cos(r, \beta))^2 + (\sin(r, \alpha) - \sin(r, \beta))^2 - 2r^2 \\
-2r \cos(r, \alpha - \beta) &= (\cos(r, \alpha) - \cos(r, \beta))^2 + (\sin(r, \alpha) - \sin(r, \beta))^2 - 2r^2
\end{aligned}$$

$$\begin{aligned}
\frac{-2r \cos(r, \alpha - \beta)}{-2r} &= \frac{(\cos(r, \alpha) - \cos(r, \beta))^2 + (\sin(r, \alpha) - \sin(r, \beta))^2 - 2r^2}{-2r} \\
\cos(r, \alpha - \beta) &= \frac{(\cos(r, \alpha) - \cos(r, \beta))^2 + (\sin(r, \alpha) - \sin(r, \beta))^2 - 2r^2}{-2r} \\
\cos(r, \alpha - \beta) &= \frac{\cos^2(r, \alpha) - 2\cos(r, \alpha)\cos(r, \beta) + \cos^2(r, \beta) + (\sin(r, \alpha) - \sin(r, \beta))^2 - 2r^2}{-2r} \\
\cos(r, \alpha - \beta) &= \frac{\cos^2(r, \alpha) - 2\cos(r, \alpha)\cos(r, \beta) + \cos^2(r, \beta) + \sin^2(r, \alpha) - 2\sin(r, \alpha)\sin(r, \beta) + \sin^2(r, \beta) - 2r^2}{-2r} \\
\cos(r, \alpha - \beta) &= \frac{(\cos^2(r, \alpha) + \sin^2(r, \alpha)) - 2\cos(r, \alpha)\cos(r, \beta) + \cos^2(r, \beta) - 2\sin(r, \alpha)\sin(r, \beta) + \sin^2(r, \beta) - 2r^2}{-2r} \\
\cos(r, \alpha - \beta) &= \frac{(\cos^2(r, \alpha) + \sin^2(r, \alpha)) - 2\cos(r, \alpha)\cos(r, \beta) + (\cos^2(r, \beta) + \sin^2(r, \beta)) - 2\sin(r, \alpha)\sin(r, \beta) - 2r^2}{-2r} \\
\cos(r, \alpha - \beta) &= \frac{r^2 - 2\cos(r, \alpha)\cos(r, \beta) + (\cos^2(r, \beta) + \sin^2(r, \beta)) - 2\sin(r, \alpha)\sin(r, \beta) - 2r^2}{-2r} \\
\cos(r, \alpha - \beta) &= \frac{r^2 - 2\cos(r, \alpha)\cos(r, \beta) + r^2 - 2\sin(r, \alpha)\sin(r, \beta) - 2r^2}{-2r} \\
\cos(r, \alpha - \beta) &= \frac{2r^2 - 2\cos(r, \alpha)\cos(r, \beta) - 2\sin(r, \alpha)\sin(r, \beta) - 2r^2}{-2r} \\
\cos(r, \alpha - \beta) &= \frac{-2\cos(r, \alpha)\cos(r, \beta) - 2\sin(r, \alpha)\sin(r, \beta)}{-2r} \\
\cos(r, \alpha - \beta) &= \frac{-2(\cos(r, \alpha)\cos(r, \beta) + \sin(r, \alpha)\sin(r, \beta))}{-2r} \\
\cos(r, \alpha - \beta) &= \frac{\cos(r, \alpha)\cos(r, \beta) + \sin(r, \alpha)\sin(r, \beta)}{r} \\
\cos(r, \alpha - \beta) &= r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\alpha - \beta}{r}\right)^{2k}}{(2k)!} \\
\cos(r, \alpha - \beta) &= r \lim_{n \rightarrow \infty} \sum_{k=0}^n \frac{(-1)^k \left(\frac{\alpha - \beta}{r}\right)^{2k}}{(2k)!} \\
\cos(r, \alpha - \beta) &= r \lim_{n \rightarrow \infty} \sum_{k=0}^n \frac{(-1)^k \sum_{j=0}^{2k} \binom{2k}{j} \left(\frac{\alpha}{r}\right)^{2k-j} \left(\frac{-\beta}{r}\right)^j}{(2k)!} \\
\cos(r, \alpha - \beta) &= r \lim_{n \rightarrow \infty} \sum_{k=0}^n \frac{(-1)^k \sum_{j=0}^{2k} \frac{(2k)!}{j!(2k-j)!} \left(\frac{\alpha}{r}\right)^{2k-j} \left(\frac{-\beta}{r}\right)^j}{(2k)!} \\
\cos(r, \alpha - \beta) &= r \lim_{n \rightarrow \infty} \sum_{k=0}^n \frac{(-1)^k \sum_{j=0}^{2k} (2k)! \frac{\left(\frac{\alpha}{r}\right)^{2k-j} \left(\frac{-\beta}{r}\right)^j}{j!(2k-j)!}}{(2k)!} \\
\cos(r, \alpha - \beta) &= r \lim_{n \rightarrow \infty} \sum_{k=0}^n \frac{(-1)^k (2k)! \sum_{j=0}^{2k} \frac{\left(\frac{\alpha}{r}\right)^{2k-j} \left(\frac{-\beta}{r}\right)^j}{j!(2k-j)!}}{(2k)!} \\
\cos(r, \alpha - \beta) &= r \lim_{n \rightarrow \infty} \sum_{k=0}^n (-1)^k \sum_{j=0}^{2k} \frac{\left(\frac{\alpha}{r}\right)^{2k-j} \left(\frac{-\beta}{r}\right)^j}{j!(2k-j)!}
\end{aligned}$$

[illegible]

0.1.3.4.1 Partial Derivative

$$\frac{\partial \sin(r, s)}{\partial s} = \frac{\partial \left(r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k+1}}{(2k+1)!} \right)}{\partial s}$$

$$\frac{\partial \sin(r, s)}{\partial s} = \frac{\partial \left(r \sum_{k=0}^{\infty} \frac{(-1)^k \frac{s^{2k+1}}{r^{2k+1}}}{(2k+1)!} \right)}{\partial s}$$

$$\frac{\partial \sin(r, s)}{\partial s} = \frac{\partial \left(r \sum_{k=0}^{\infty} \frac{(-1)^k s^{2k+1}}{r^{2k+1} (2k+1)!} \right)}{\partial s}$$

$$\frac{\partial \sin(r, s)}{\partial s} = \frac{\partial \left(\sum_{k=0}^{\infty} \frac{(-1)^k r s^{2k+1}}{r^{2k+1} (2k+1)!} \right)}{\partial s}$$

$$\frac{\partial \sin(r, s)}{\partial s} = \frac{\partial \left(\sum_{k=0}^{\infty} \frac{(-1)^k s^{2k+1}}{r^{2k} (2k+1)!} \right)}{\partial s}$$

$$\frac{\partial \sin(r, s)}{\partial s} = \sum_{k=0}^{\infty} \frac{(-1)^k \frac{\partial s^{2k+1}}{\partial s}}{r^{2k} (2k+1)!}$$

$$\frac{\partial \sin(r, s)}{\partial s} = \sum_{k=0}^{\infty} \frac{(-1)^k (2k+1) s^{2k}}{r^{2k} (2k+1)!}$$

$$\frac{\partial \sin(r, s)}{\partial s} = \sum_{k=0}^{\infty} \frac{(-1)^k (2k+1) \frac{s^{2k}}{r^{2k}}}{(2k+1)!}$$

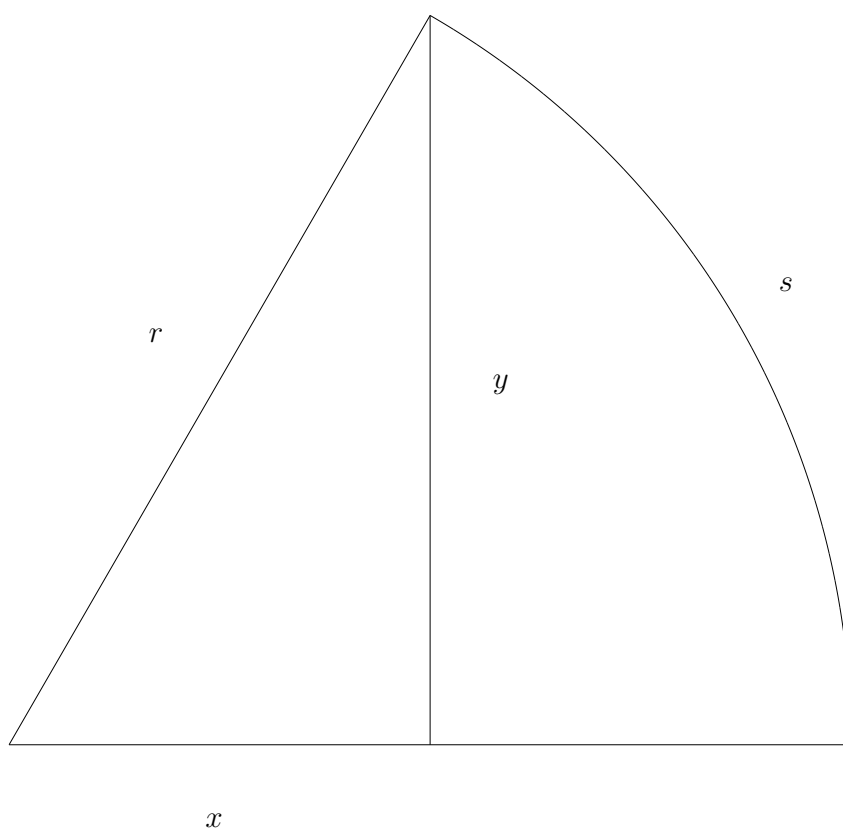
$$\frac{\partial \sin(r, s)}{\partial s} = \sum_{k=0}^{\infty} \frac{(-1)^k (2k+1) \left(\frac{s}{r}\right)^{2k}}{(2k+1)!}$$

$$\frac{\partial \sin(r, s)}{\partial s} = \sum_{k=0}^{\infty} \frac{(-1)^k (2k+1) \left(\frac{s}{r}\right)^{2k}}{(2k+1)(2k)!}$$

$$\frac{\partial \sin(r, s)}{\partial s} = \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!}$$

$$\frac{\partial \sin(r, s)}{\partial s} = \frac{r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!}}{r}$$

$$\frac{\partial \sin(r, s)}{\partial s} = \frac{\cos(r, s)}{r}$$



0.2 Circular Sector Sequence

0.2.1 Apothem

$$\begin{aligned}
A_{ORS_{n-1}} &= \frac{\overline{RS_{n-1}OX_n}}{2}, n \geq 1 \\
A_{ORS_{n-1}} &= \frac{\overline{OS_{n-1}RX_{n-1}}}{2}, n \geq 1 \\
\frac{\overline{RS_{n-1}OX_n}}{2} &= \frac{\overline{OS_{n-1}RX_{n-1}}}{2}, n \geq 1 \\
\overline{RS_{n-1}OX_n} &= \overline{OS_{n-1}RX_{n-1}}, n \geq 1 \\
\frac{\overline{RS_{n-1}OX_n}}{\overline{RS_{n-1}}} &= \frac{\overline{OS_{n-1}RX_{n-1}}}{\overline{RS_{n-1}}}, n \geq 1 \\
\overline{OX_n} &= \frac{\overline{OS_{n-1}RX_{n-1}}}{\overline{RS_{n-1}}}, n \geq 1 \\
\overline{OX_n} &= \frac{\overline{OS_{n-1}}\sqrt{\overline{OR}^2 - \overline{OX_{n-1}}^2}}{\overline{RS_{n-1}}}, n \geq 1 \\
\overline{OX_n} &= \frac{\overline{OS_{n-1}}\sqrt{\overline{OR}^2 - \overline{OX_{n-1}}^2}}{\sqrt{\overline{OR}^2 + \overline{OS_{n-1}}^2 - 2\overline{OS_{n-1}}\overline{OX_{n-1}}}}, n \geq 1
\end{aligned}$$

0.2.2 Chord Length

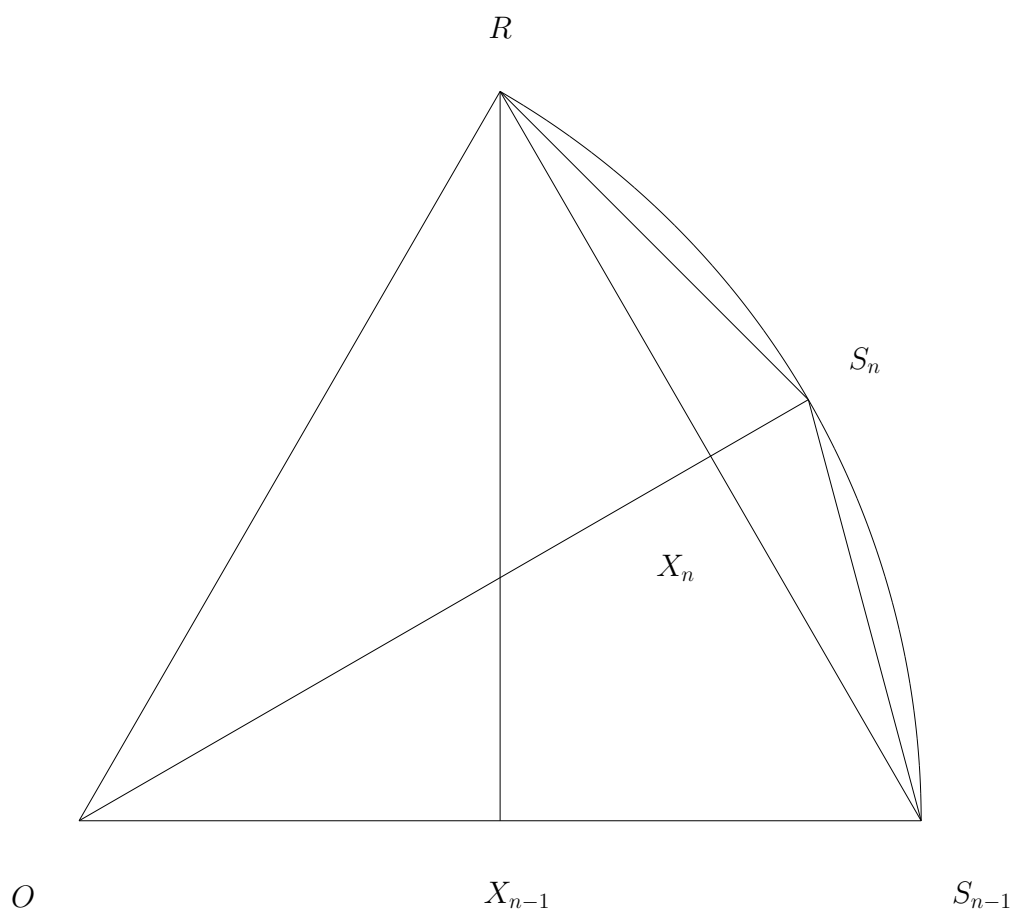
$$\begin{aligned}
\overline{RS_n} &= \sqrt{\overline{RX_n}^2 + \overline{X_nS_n}^2} \\
\overline{RS_n} &= \sqrt{\overline{OR}^2 - \overline{OX_n}^2 + \overline{X_nS_n}^2} \\
\overline{RS_n} &= \sqrt{\overline{OR}^2 - \overline{OX_n}^2 + (\overline{OS_n} - \overline{OX_n})^2} \\
\overline{RS_n} &= \sqrt{\overline{OR}^2 - \overline{OX_n}^2 + \overline{OS_n}^2 - 2\overline{OS_n}\overline{OX_n} + \overline{OX_n}^2} \\
\overline{RS_n} &= \sqrt{\overline{OR}^2 + \overline{OS_n}^2 - 2\overline{OS_n}\overline{OX_n}}
\end{aligned}$$

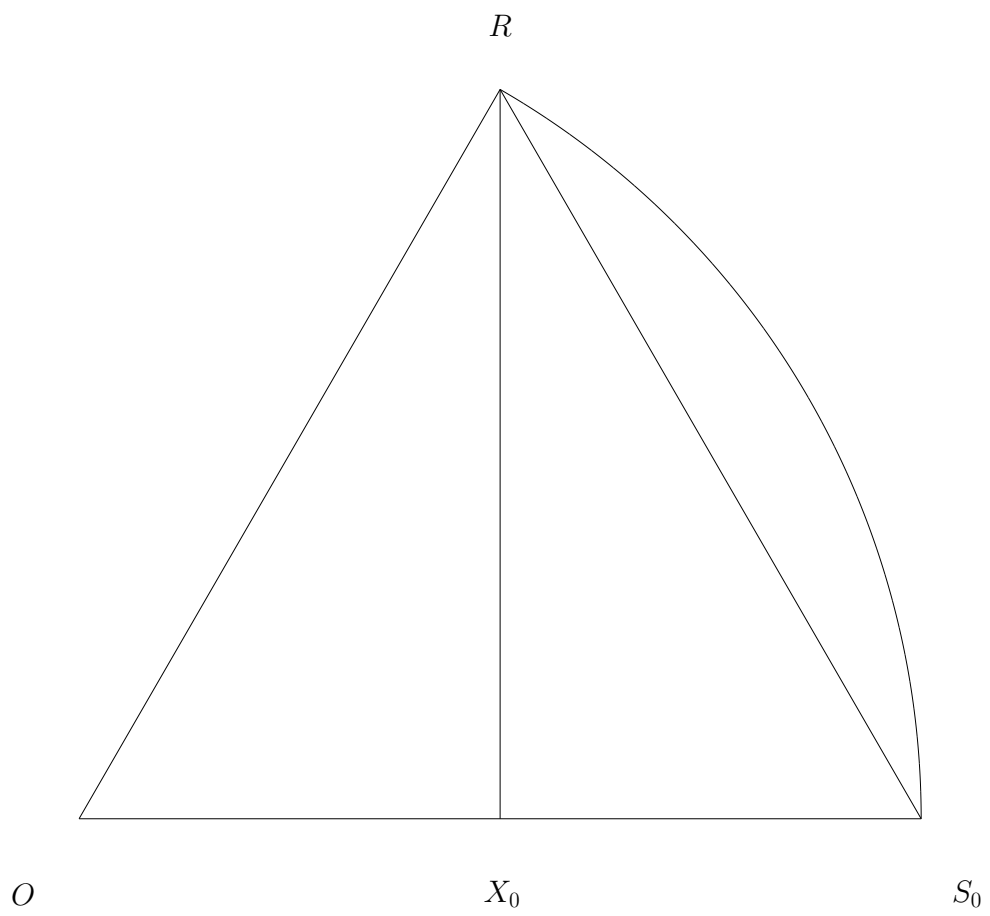
0.2.3 Arc Length

$$\begin{aligned}
\text{arc } RS_0 &= \lim_{n \rightarrow \infty} 2^n \overline{RS_n} \\
\text{arc } RS_0 &= \lim_{n \rightarrow \infty} 2^n \sqrt{\overline{OR}^2 + \overline{OS_n}^2 - 2\overline{OS_n}\overline{OX_n}}
\end{aligned}$$

0.2.4 Area

$$\begin{aligned}
A_0 &= \lim_{n \rightarrow \infty} 2^{n-1} A_{ORS_n S_{n-1}} \\
A_0 &= \lim_{n \rightarrow \infty} 2^{n-1} \frac{\overline{RS_{n-1} OS_n}}{2} \\
A_0 &= \frac{\lim_{n \rightarrow \infty} 2^{n-1} \overline{RS_{n-1} OS_n}}{2} \\
A_0 &= \frac{\lim_{n \rightarrow \infty} (2^{n-1} \overline{RS_{n-1}}) \overline{OS_n}}{2} \\
A_0 &= \frac{\lim_{n \rightarrow \infty} \overline{OS_n} (2^{n-1} \overline{RS_{n-1}})}{2} \\
A_0 &= \frac{\left(\lim_{n \rightarrow \infty} \overline{OS_n} \right) \left(\lim_{n \rightarrow \infty} 2^{n-1} \overline{RS_{n-1}} \right)}{2} \\
A_0 &= \frac{\overline{OR} \lim_{n \rightarrow \infty} 2^{n-1} \overline{RS_{n-1}}}{2} \\
A_0 &= \frac{\overline{OR} \text{arc } RS_0}{2}
\end{aligned}$$





0.3 Simple Harmonic Motion

$$[m] = \text{kg}$$

$$[\omega] = \text{m/s}$$

$$[k] = \text{N/m}$$

$$[r] = \text{m}$$

$$F = ma$$

$$F = -kx$$

$$ma = -kx$$

$$x(t) = \cos(r, \omega t)$$

$$a(t) = \frac{d^2 x}{dt^2}$$

$$a(t) = \frac{d^2 \cos(r, \omega t)}{dt^2}$$

$$a(t) = \frac{-\omega^2 \cos(r, \omega t)}{r^2}$$

$$a(x) = \frac{-\omega^2 x}{r^2}$$

$$m \frac{-\omega^2 x}{r^2} = -kx$$

$$\frac{m(-\omega^2 x)}{r^2} = -kx$$

$$\frac{-m\omega^2 x}{r^2} = -kx$$

$$\frac{-m\omega^2 x}{r^2} = \frac{-kr^2 x}{r^2}$$

$$m\omega^2 = kr^2$$

$$v(t) = \frac{dx}{dt}$$

$$v(t) = \frac{d \cos(r, \omega t)}{dt}$$

$$v(t) = \frac{-\omega \sin(r, \omega t)}{r}$$

$$K(t) = \frac{mv^2(t)}{2}$$

$$K(t) = \frac{m \left(\frac{-\omega \sin(r, \omega t)}{r} \right)^2}{2}$$

$$K(t) = \frac{m \frac{(-\omega \sin(r, \omega t))^2}{r^2}}{2}$$

$$K(t) = \frac{m \frac{(-\omega)^2 \sin^2(r, \omega t)}{r^2}}{2}$$

$$K(t) = \frac{m \frac{\omega^2 \sin^2(r, \omega t)}{r^2}}{2}$$

$$K(t) = \frac{\left(\frac{m\omega^2 \sin^2(r, \omega t)}{r^2} \right)}{2}$$

$$K(t) = \frac{\left(\frac{kr^2 \sin^2(r, \omega t)}{r^2}\right)}{2}$$

$$K(t) = \frac{k \sin^2(r, \omega t)}{2}$$

$$U(t) = \frac{kx^2(t)}{2}$$

$$U(t) = \frac{k \cos^2(r, \omega t)}{2}$$

$$E = K + U$$

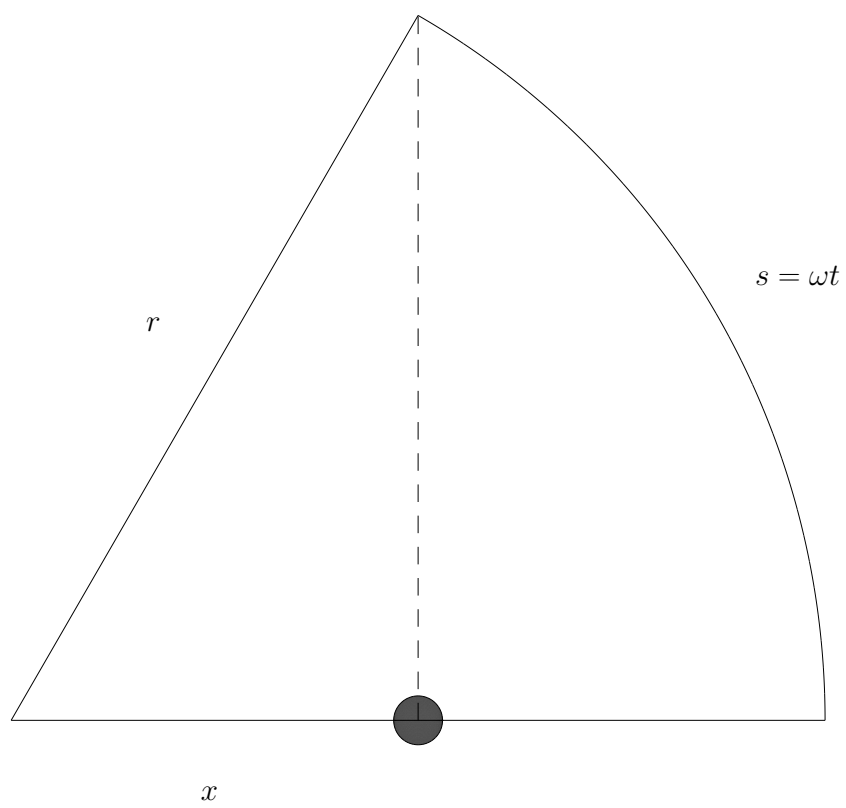
$$E = \frac{k \sin^2(r, \omega t)}{2} + \frac{k \cos^2(r, \omega t)}{2}$$

$$E = \frac{k \sin^2(r, \omega t) + k \cos^2(r, \omega t)}{2}$$

$$E = \frac{k(\sin^2(r, \omega t) + \cos^2(r, \omega t))}{2}$$

$$E = \frac{kr^2}{2}$$

$$E = \frac{m\omega^2}{2}$$



0.4 Uniform Circular Motion

$$[m] = \text{kg}$$

$$[\omega] = \text{m/s}$$

$$[r] = \text{m}$$

$$s = \omega t$$

$$\omega = \frac{\Delta s}{\Delta t}$$

$$x(t) = \cos(r, \omega t)$$

$$y(t) = \sin(r, \omega t)$$

$$\vec{r} = (x, y)$$

$$\vec{r}(t) = (\cos(r, \omega t), \sin(r, \omega t))$$

$$\vec{v} = \frac{d\vec{r}}{dt}$$

$$\vec{v}(t) = \frac{d(\cos(r, \omega t), \sin(r, \omega t))}{dt}$$

$$\vec{v}(t) = \left(\frac{d \cos(r, \omega t)}{dt}, \frac{d \sin(r, \omega t)}{dt} \right)$$

$$\vec{v}(t) = \left(\frac{-\omega \sin(r, \omega t)}{r}, \frac{\omega \cos(r, \omega t)}{r} \right)$$

$$\vec{v}(t) = \frac{\omega (-\sin(r, \omega t), \cos(r, \omega t))}{r}$$

$$v = |\vec{v}|$$

$$v = \left| \frac{\omega (-\sin(r, \omega t), \cos(r, \omega t))}{r} \right|$$

$$v = \frac{|\omega (-\sin(r, \omega t), \cos(r, \omega t))|}{|r|}$$

$$v = \frac{|\omega| |(-\sin(r, \omega t), \cos(r, \omega t))|}{|r|}$$

$$v = \frac{\omega |(-\sin(r, \omega t), \cos(r, \omega t))|}{|r|}$$

$$v = \frac{\omega \sqrt{(-\sin(r, \omega t))^2 + \cos^2(r, \omega t)}}{|r|}$$

$$v = \frac{\omega \sqrt{\sin^2(r, \omega t) + \cos^2(r, \omega t)}}{|r|}$$

$$v = \frac{\omega \sqrt{r^2}}{|r|}$$

$$v = \frac{\omega |r|}{|r|}$$

$$v = \omega$$

$$E = \frac{mv^2}{2}$$

$$E = \frac{m\omega^2}{2}$$

$$\vec{a} = \frac{d^2\vec{r}}{dt^2}$$

$$\vec{a}(t) = \frac{d^2(\cos(r, \omega t), \sin(r, \omega t))}{dt^2}$$

$$\vec{a}(t) = \left(\frac{d^2 \cos(r, \omega t)}{dt^2}, \frac{d^2 \sin(r, \omega t)}{dt^2} \right)$$

$$\vec{a}(t) = \left(\frac{-\omega^2 \cos(r, \omega t)}{r^2}, \frac{-\omega^2 \sin(r, \omega t)}{r^2} \right)$$

$$\vec{a}(t) = \frac{-\omega^2 (\cos(r, \omega t), \sin(r, \omega t))}{r^2}$$

$$\vec{a} = \frac{-\omega^2 \vec{r}}{r^2}$$

$$a = |\vec{a}|$$

$$a = \left| \frac{-\omega^2 \vec{r}}{r^2} \right|$$

$$a = \frac{|-\omega^2 \vec{r}|}{|r^2|}$$

$$a = \frac{|-\omega^2| |\vec{r}|}{|r^2|}$$

$$a = \frac{\omega^2 |\vec{r}|}{|r^2|}$$

$$a = \frac{\omega^2 r}{|r^2|}$$

$$a = \frac{\omega^2 r}{r^2}$$

$$a = \frac{\omega^2}{r}$$

$$F = ma$$

$$F = m \frac{\omega^2}{r}$$

$$F = \frac{m\omega^2}{r}$$

